FREEDOM INTERNATIONAL SCHOOL
L-2 ELECTROCHEMISTRY
WORKSHEET

1.	Conductivity of 2.5×10^{-4} M methanoic acid is 5.25×10^{-5} S cm^{-1} Calculate its molar conductivity and degree of dissociation. Given: $\Lambda°(H^+) = 349.5$ S cm^2 mol^{-1} and $\Lambda°(HCOO^-) = 50.5$ S cm^2 mol^{-1}
2.	The conductivity of 0.20 mol L^{-1} solution of KCl is 2.48×10^{-2} S cm^{-1}. Calculate its molar conductivity and degree of dissociation (a). Given $\Lambda°(K+) = 73.5$ S cm^2 mol^{-1} and $\Lambda°(Cl-) = 76.5$ S cm^2 mol^{-1}.
3.	The conductivity of 0.1 mol L^{-1} solution of NaCl is 1.06×10^{-2} S cm^{-1}. Calculate its molar conductivity and degree of dissociation. Given $\Lambda°_{(Na+)} = 50.1$ S $cm^2 mol^{-1}$ and $\Lambda°_{(C1-)} = 76.5$ S cm^2 mol^{-1}
4.	The resistance of 0.01 M NaCl solution at 25 °C is 200 Ω. The cell constant of conductivity cell is unity. Calculate the molar conductance.
5.	Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100 ohms. If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520 ohms, calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} is 1.29×10^{-2} S cm^{-}
6.	The conductivity of 0.20 M solution of KCl at 298 K is 0.025 S cm^{-1}. Calculate its molar conductivity.
7.	The resistance of a conductivity cell containing 10^{-3} M KCl solution at 25 °C is 1500 ohms. What is the cell constant if conductivity of 10^{-3} M KCl solution at 25 °C is 1.5×10^{-4} S cm^{-1}?
8.	The molar conductivity of a 1.5 M solution of an electrolyte is found to be 138.9 S cm^2 mol^{-1}. Calculate the conductivity of the solution.

9.	The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is 5.55 X 10^3 ohm. Calculate its resistivity, conductivity and molar conductivity.
10.	When a certain electrolytic cell was filled with 0.1 M KCl, it has resistance of 85 ohms at 25 °C. When the same cell was filled with an aqueous solution of 0.052 M unknown electrolyte, the resistance was 96 ohms. Calculate the molar conductance of the electrolyte at this concentration. [Specific conductance of 0.1 M KCl = 1.29 X 10^{-2}ohm^{-1} cm^{-1}]
11.	Calculate the degree of dissociation of acetic acid at 298 K, given that: Λ_m (CH_3COOH) = 11.7 S cm^2 mol^{-1}, $\Lambda°_m$ (CH_3COO) = 40.9 S cm^2 mol^{-1}, $\Lambda°(H^+)$ = 349.1 S cm^2 mol^{-1}.
12.	Conductivity of 0.01 M NaCl solution is 0.12 Sm^{-1} . Calculate its molar conductivity.
13.	At 298 K, molar conductance of 0.1M aqueous solution of ammonium hydroxide is 9.54 Scm^{2}mol^{-1} and at infinite dilution its molar conductance is 238 S cm^{2}mol^{-1}. Calculate the degree of ionization of ammonium hydroxide at the same concentration and temperature.
14.	The equivalent conductance of M/32 solution of a weak monobasic acid is 8.0 S cm^2 eq^{-1} and equivalent conductance at infinite dilution is 400 S cm^2 eq^{-1}. What is the dissociation constant of the acid?

15.	Calculate E°cell for the following reaction at 25 °C: $A + B^{2+}$ (0.001 M) → A^{2+} (0.0001 M) + B Given : E_{cell} = 2.6805 V, 1 F = 96500 C mol^{-1}
16.	Calculate emf of the following cell at 25 °C: Fe | Fe^{2+}(0.001 M) ‖ H^{+}(0.01 M) | H_2(g) (1 bar) | Pt(s) $E°_{(Fe2+\|Fe)}$ = -0.44V, $E°_{(H+\|H2)}$ = 0.00 V
17.	Calculate the emf of the following cell at 25 °C: Zn | Zn^{2+} (0.001 M) | | H^{+}(0.01 M) | H_2(g) (1 bar) | Pt(s)
18.	Calculate ΔG° for the reaction: Mg(s) + Cu^{2+}(aq) → Mg^{2+}(aq) + Cu(s) Given: $E°_{cell}$ = + 2.71 V, 1 F = 96500 C mol^{-1}
19.	Equilibrium constant (Kc) for the given cell reaction is 10. Calculate $E°_{cell}$. $A_{(s)} + B^{+2}_{(aq)} \rightarrow A^{2+}_{(aq)} + B_{(s)}$
20.	Calculate emf of the following cell at 298 K: Mg(s) | Mg^{2+}(0.1 M) ‖ Cu^{2+} (0.01) | Cu(s). [Given $E°_{cell}$, = + 2.71 V, 1 F = 96500 C mol^{-1}]
21.	The standard electrode potential (E°) for Daniel cell is + 1.1 V. Calculate the ΔG° and equilibrium constant for the same. (Given: antilog 0.225 = 1.67)
22.	Calculate the emf of the following cell at 25 °C: Ag(s) | Ag^{+} (10^{-3} M) ‖ Cu^{2+} (10^{-1} M) | Cu(s). Given: $E°_{Cu+2/Cu}$ = + 0.34 V, $E°_{Ag+/Ag}$ = + 0.8 V
23.	A zinc rod is dipped in 0.1 M solution of $ZnSO_4$. The salt is 95% dissociated at this dilution at 298 K. Calculate the electrode potential. (Given log 95 = 1.977) Given $E^{0}_{(Zn2+/Zn)}$ = – 0.76 V
24.	A voltaic cell is set up at 25 °C with the following half cells: Al/Al^{3+}(0.001 M) and Ni/Ni^{2+}(0.50 M). Write an equation for the reaction that occurs when the cell generates an electric current and determine the cell potential. [$E^{0}_{(Ni2+/Ni)}$= -0.25 V and $E^{0}_{(Al3+/Al)}$ = – 1.66 V](Given: log 2 = 0.3010, log 5 = 0.6991)
25.	Calculate the cell potential, E, at 25 °C for the cell if the initial concentration of $Ni(NO_3)_2$ is 0.100 molar and the initial concentration of $AgNO_3$ is 1.00 molar. [$E^{0}_{(Ni2+/Ni)}$= -0.25 V; $E^{0}_{(Ag+/Ag)}$ = 0.80 V]
26.	Determine the values of equilibrium constant (Kc) and ΔG° for the following reaction: Ni(s) + $2Ag^{+}$(aq) → Ni^{+2}(aq) + 2Ag(s), $E°_{cell}$ = 1.05 V Given: antilog 0.5329 = 3.411
27.	Two half-reactions of an electrochemical cell are given below: MnO_4^{-}(aq) + $8H^{+}$(aq) + $5e^{-}$ → Mn^{2+}(aq) + $4H_2O$(l), E° = + 1.51 V Sn^{2+}(aq) → Sn^{4+}(aq) + $2e^{-}$, E° = + 0.15 V

	Construct a redox equation from the standard potential of the cell and predict whether the reaction is reactant favoured or product favoured. Calculate the free energy change.
28.	For the cell Zn(s) | Zn^{2+}(2M) | | Cu^{2+}(0.5 M) | Cu(s) (a) Write equation for each half-reaction. (b) Calculate the cell potential at 25°C. [$E^o_{(Zn2+/Zn)}$= – 0.76 V; $E^o_{(Cu2+/Cu)}$= + 0 .34V]](Given: log 2 = 0.3010, log 4 = 0.6021, log 5 = 0.6991)
29.	Calculate the potential for half-cell containing 0.10 M $K_2Cr_2O_7$(aq), 0.20 M Cr^{3+}(aq) and 1.0x10^4M H^+(aq). The half-cell reaction is $K_2Cr_2O_7$ (aq) + 14 H^+(aq) + 6e- → 2Cr^{3+}(aq) + 7$H_2$0(l) and the standard electrode potential is given as E°= 1.33 V.
30.	Calculate the equilibrium constant for the equilibrium reaction: Fe(s) + Cd^{+2} (aq) → Fe^{2+} (aq) + Cd(s) ($E^o_{(Cd2+/Cd)}$= -0.40 V, $E^o_{(Fe2+/Fe)}$= -0.44 V).
31.	A copper-silver cell is set up. The copper ion concentration is 0.10 M. The concentration of silver ion is not known. The cell potential when measured was 0.422 V. Determine the concentration of silver ions in the cell. (Given$E^0_{(Ag+/Ag)}$= +0.80V, $E^0_{(Cu2+/Cu)}$ = +0.34V
32.	Calculate the standard free energy change for the following reaction at 298 K. $Au(s) + Ca^{+2}(1M) \rightarrow Au^{+3}(1M) + Ca(s)$ $E^o_{Au^{+3}/Au} = +1.50V, E^o_{Ca^{+2}/Ca} = -2.87V$ Predict whether the reaction will be spontaneous or not at 298K. Which one of the above two half-cells will act as an oxidizing agent and reducing agent?
33.	Write the Nernst equation and calculate emf of the following cells at 298 K. a) Sn (s) | Sn^{+2} (0.05M) ‖ H^+ (0.02M) | H_2 (1bar) | Pt b) Pt(s) | Br_2 (l) | Br^-(0.01M) ‖ H^+ (0.03M) | H_2(1bar) | Pt Given: $E^o_{Sn+2/Sn} = -0.14\,V, E^o_{\frac{Br2}{Br}-} = +1.08V$